Data-Driven Electric Vehicle Charging Optimization Using Real-World ACN Dataset

Discrete Optimization

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1 Problem Description

1.1 Problem Situation (Business Case)

As electric vehicle (EV) adoption grows, workplace and campus charging facilities face increasing pressure from concurrent charging demand. When a large number of vehicles charge simultaneously, the resulting power spike-referred to as peak demand-can lead to high electricity costs due to demand charges imposed by utilities.

In practice, commercial electricity tariffs consist of two components: **Time-of-Use (TOU) energy charges**, which vary by time of day, and **demand charges**, which are billed based on the highest power draw recorded during a billing period. Both components incentivize load shifting and peak reduction, making uncontrolled charging-where each vehicle charges at full rate upon arrival-a costly baseline strategy. In the case of the Caltech campus, electricity is supplied under Southern California Edison (SCE) commercial tariffs, which include both TOU energy charges and facilities-related demand (FRD) charges [5, 1].

The **Caltech Adaptive Charging Network (ACN)** provides a publicly available dataset of workplace EV charging sessions collected from over 55 charging stations on the Caltech campus [2, 4]. Each session record includes the vehicle's connection time, disconnection time, and energy delivered. Lee et al. [3] described the architecture of the ACN system and presented a convex optimization-based scheduling algorithm under this infrastructure, reporting that coordinated scheduling reduces operating costs relative to uncontrolled charging.

The EV charging scheduling problem involves two interrelated objectives: minimizing the total charging cost under TOU pricing and minimizing the peak load. Wu et al. [6] analyzed the structural properties of these two objectives, showing that cost minimization under TOU pricing exhibits a greedy-choice property, while peak load minimization does not. This distinction motivates the use of mathematical programming approaches rather than simple heuristics for combined cost and peak optimization.

This project formulates the scheduling problem as a **Mixed Integer Programming model (MIP)** that minimizes peak power demand subject to operational constraints. Binary decision variables for charging on/off states are introduced, yielding a discrete optimization problem solvable by commercial MIP solvers.

1.2 Model Formulation

1.2.1 Input

The following inputs define each scheduling instance, sourced from the Caltech ACN-Dataset [2, 4]:

Table 1.1: Caltech ACN-Data

Parameter	Symbol	Description
Vehicle set	$i \in I$	Set of EVs present in the scheduling horizon
Time slot set	$t \in T$	Discretized time slots (e.g., 15-min intervals)
Arrival time	a_i	Time slot when vehicle i connects to charger
Departure time	d_i	Time slot when vehicle i must disconnect
Required energy	e_i (kWh)	Total energy to be delivered to vehicle i before departure
Max charging rate	$\bar{r}_i$ (kW)	Maximum power output of charger assigned to vehicle i
Transformer capacity	C (kW)	Aggregate power capacity of the site transformer
TOU price	λ_t (\$/kWh)	Time-varying electricity price at time slot t
Demand charge rate	μ (\$/kW)	Charge per kW of peak power during billing period
Min charging rate	$\underline{r_i}$ (kW)	Minimum power threshold below which charger cannot operate (optional)
Max interruptions	K_i	Upper bound on number of charging interruptions allowed per vehicle (optional)

1.2.2 Modeling Assumptions

The current formulation is based on the following assumptions. These may be revisited or relaxed as the project progresses, depending on experimental findings:

- Each vehicle is pre-assigned to a specific charger (i.e., the charger allocation problem is not addressed in this model).
- Scheduling is performed offline with full knowledge of arrival/departure times and energy requirements (deterministic setting).

- The objective is simplified to minimize the peak demand only (min Z). Although the SCE TOU-EV-8 tariff is available through the acnportal library, its demand charge is zero ($\mu = 0$), rendering the demand charge term inactive. TOU-based cost minimization is reserved as a future extension.
- Practical constraints such as minimum charging thresholds (C7) and limits on charging interruptions (C8) are included as optional extensions. Their activation will be determined based on initial experimental observations.

1.2.3 Decision Variables

The model uses the following decision variables:

$r_{it} \geq 0$ **(continuous)** : Charging rate (kW) for vehicle i at time slot t.

$y_{it} \in \{0,1\}$ **(binary)** : Equals 1 if vehicle i is actively charged at time slot t, and 0 otherwise.

$Z \geq 0$ **(continuous, auxiliary)** : Peak power demand (kW) across all time slots.

1.2.4 Objective Function

The objective minimizes the total electricity cost, comprising TOU energy cost and the demand charge:

$$min \sum_t (\lambda_t \cdot \sum_i r_{it} \cdot \Delta t) + \mu \cdot Z$$

The first term captures time-varying energy costs: shifting load to off-peak periods (lower λ_t) reduces this component. The second term penalizes the peak power through the demand charge coefficient μ, which encourages load flattening across the scheduling horizon. As noted by Wu et al. [6], cost minimization under TOU pricing admits a greedy-choice property in continuous relaxation, while peak minimization requires global coordination—combining both in a single objective with binary variables requires an integer programming approach.

Why min Z (Peak Minimization Only)

In this study, since the applicable SCE TOU-EV-8 tariff carries no demand charge ($\mu = 0$), the objective is simplified to:

$$\min Z$$

The energy cost minimization of the TOU is reserved as a future extension.

1.2.5 Constraints

C1. Energy fulfillment: $\sum_t r_{it} \cdot \Delta t \geq e_i \ , \forall i \in I$

Each vehicle must receive its required energy before departure.

C2. On/off coupling: $r_{it} \leq \bar{r}_i \cdot y_{it}$, $\forall i, t$

Charging rate is zero when $y_{it} = 0$ and bounded by the charger's rated capacity when $y_{it} = 1$.

C3. Availability window: $y_{it} = 0$, $\forall t < a_i$ or $t > d_i$

Charging is restricted to the interval during which the vehicle is physically connected.

C4. Peak demand definition: $\sum_i r_{it} \leq Z$, $\forall t \in T$

Z represents the maximum aggregate power across all time slots.

C5. Transformer capacity: $\sum_i r_{it} \leq C$, $\forall t \in T$

Aggregate power must not exceed the site's transformer rating [3].

C6. Variable domains: $r_{it} \geq 0$, $y_{it} \in \{0, 1\}$, $Z \geq 0$

Practical Extension Constraints

The following constraints address the operational realities observed in EV charging infrastructure. They are included in the formulation as optional extensions and may be activated based on preliminary experimental findings.

C7. Minimum charging rate: $r_{it} \geq \underline{r}_i \cdot y_{it}$, $\forall i, t$

When a charger is active ($y_{it} = 1$), the charging rate must meet a minimum hardware threshold. Most commercial EV chargers cannot operate below a certain power level; this constraint prevents the model from assigning impractically small charging rates.

C8. Charging interruption limit: $\sum_t (y_{i,t-1} - y_{it})^+ \leq K_i$, $\forall i \in I$

The number of transitions from charging to idle within a vehicle's session is bounded by K_i. Frequent on/off cycling may affect battery health and user experience. The $()^+$ operator denotes the positive part, counting only stop events.

Note: Including C7 and C8 preserves the MIP structure of the model. Their impact on solution quality and computational cost will be assessed during the validation phase to determine whether they should be retained in full-scale experiments.

1.2.6 Output

The model produces the following outputs:

- **Optimal charging schedule** (r^*_{it}, y^*_{it}): Charging rate (kW) and state of on/off per vehicle per time slot.
- **Peak power** Z^* (kW): Minimized peak power demand in all time slots.

- **Energy fulfillment verification**: Confirmation that each vehicle receives its required energy e_i before departure.
- **Load profiles**: Time-series power consumption curves that compare the optimal MIP solution with UC, EDF, and RR baselines.
- **Gantt chart**: Per-vehicle charging timeline visualizing the optimal schedule.
- **Performance comparison table**: Peak power, energy fulfillment rate and computation time across all four methods (MIP, UC, EDF, RR).
- **Solver statistics**: Gurobi computation time and optimality gap.

2 Solution Approach

2.1 MIP-Based Optimization

2.1.1 Solver Configuration (Gurobi)

Gurobi is selected as the MIP solver due to its robust support for mixed-integer programming and its wide adoption in operations research literature. The model is solved using Gurobi Optimizer version 13.0.1 under an academic license. The solver is configured with default parameters and a solution is accepted only when the solver status is optimal, guaranteeing global optimality.

2.1.2 Implementation Details

The model is implemented in Python using the `gurobipy` interface. Each scheduling instance corresponds to a single day of charging sessions extracted from the ACN dataset. Time is discretized into 15-minutes ($\Delta t = 0.25$ h), yielding $|T| = 96$ slots per day. To reduce model size, decision variables r_{it} and y_{it} are only generated for slots within each vehicle's availability window $[a_i, d_i]$; variables outside this range are implicitly fixed to zero.

2.2 Baseline Strategy

To evaluate the effectiveness of the MIP-based schedule, three benchmark methods are implemented in the same instance.

2.2.1 Uncontrolled Charging (UC)

Each vehicle begins charging at its maximum rate $\bar{r}_i$ immediately upon arrival and continues until its energy requirement e_i is met or it departs. This greedy, first-come-first-served strategy represents the default behavior when no centralized scheduling is applied and serves as the worst-case baseline.

2.2.2 Earliest Deadline First (EDF)

In each time slot, active vehicles are sorted in ascending order of departure time d_i. Each vehicle is assigned its maximum charging rate in priority order until the transformer capacity C is exhausted. This

heuristic mimics deadline-aware scheduling policies used in real-time systems and is computationally trivial ($O(n \log n)$ per slot).

2.2.3 Round-Robin (RR)

At each time slot, the available transformer capacity C is distributed equally among all currently connected vehicles. Each vehicle receives a share of $C/|A_t|$, where A_t denotes the set of active vehicles in slot t, capped by its individual charger limit $\bar{r}_i$. This heuristic naturally spreads load over time and serves as a simple load-balancing reference ($O(n)$ per slot).

3 Experimental Setup

3.1 Dataset Description

The experiments use data from the real-world EV charging session from the Caltech Adaptive Charging Network (ACN), publicly available via the ACN-Data API [2]. In this study, two sites are used:

Table 3.1: ACN Dataset Summary

Site	Period	Sessions	Avg. Vehicles/Day	Transformer Cap.
office001	2019-03-25 $\sim$ 2021-09-13	1,659	2.4	50.0 kW
Caltech	2018-04-25 $\sim$ 2021-09-13	30,721	28.4	150.0 kW

3.2 Data Preprocessing

Raw session data undergoes the following preprocessing steps before being used for optimization: (1) conversion of timestamp columns to local timezone (America/Los_Angeles) to ensure consistent time representation throughout sessions, (2) removal of sessions with stay duration shorter than 15 minutes or longer than 24 hours as infeasible outliers, and (3) discretization of continuous timestamps into 15 minute time slots ($\Delta t = 0.25$ h), resulting in arrival slot a_i and departure slot d_i per session.

Table 3.2 summarizes the effect of preprocessing on each dataset.

Table 3.2: Preprocessing Summary

Site	Criterion	Before	After	Removed
office001	Total sessions	1,683	1,660	23
	Duration $<$ 15 min	–	–	3
	Duration $>$ 24 hr	–	–	20
	$kWh \leq 0$	–	–	0
Caltech	Total sessions	31,399	30,721	678
	Duration $<$ 15 min	–	–	419
	Duration $>$ 24 hr	–	–	259
	$kWh \leq 0$	–	–	0

Table 3.3 presents descriptive statistics of key attributes of the session before and after preprocessing.

Table 3.3: Descriptive Statistics Before and After Preprocessing

Site	Attribute	Mean		Std	
		Before	After	Before	After
office001	Duration (hr)	6.42	6.03	4.79	2.73
	Energy (kWh)	17.27	16.99	12.01	11.54
Caltech	Duration (hr)	5.65	5.38	5.97	3.83
	Energy (kWh)	9.25	9.30	8.75	8.69

Figures 3.1–3.3 illustrate the distribution of key session attributes before and after preprocessing for both sites.

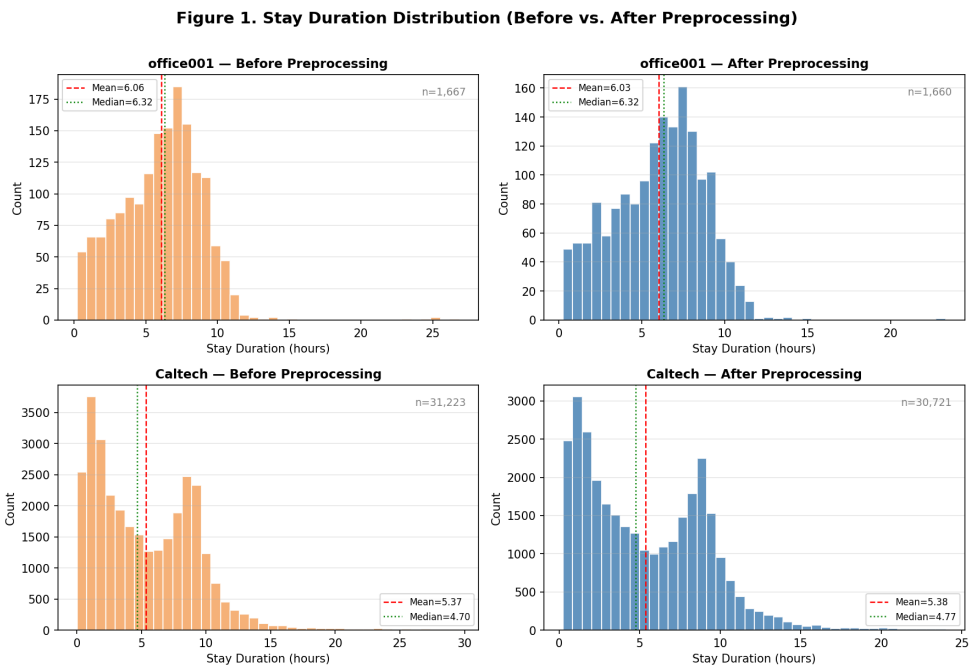

Figure 3.1: Stay Duration Distribution before and after preprocessing for office001 and Caltech. For office001, the distribution shape remains largely unchanged before and after preprocessing, as the site contains relatively few sessions with little outlier presence. For Caltech, a noticeable reduction in session count is observed after preprocessing, as reflected in the y-axis scale.

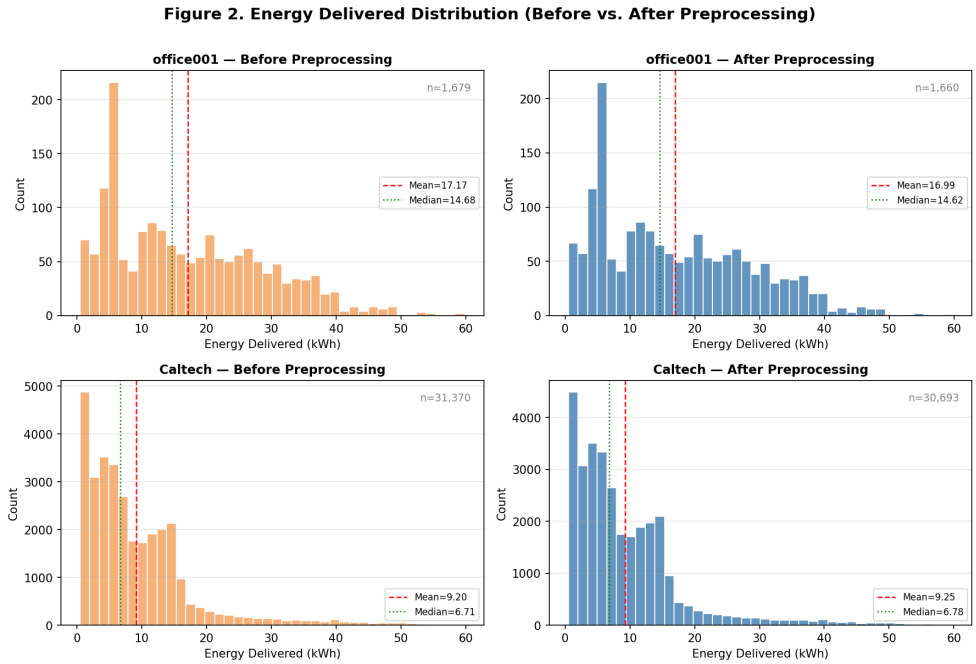

Figure 3.2: Energy Delivered Distribution before and after preprocessing for office001 and Caltech. As with the duration distribution, office001 shows no significant change in distribution shape, while Caltech exhibits a visible reduction in session count following outlier removal.

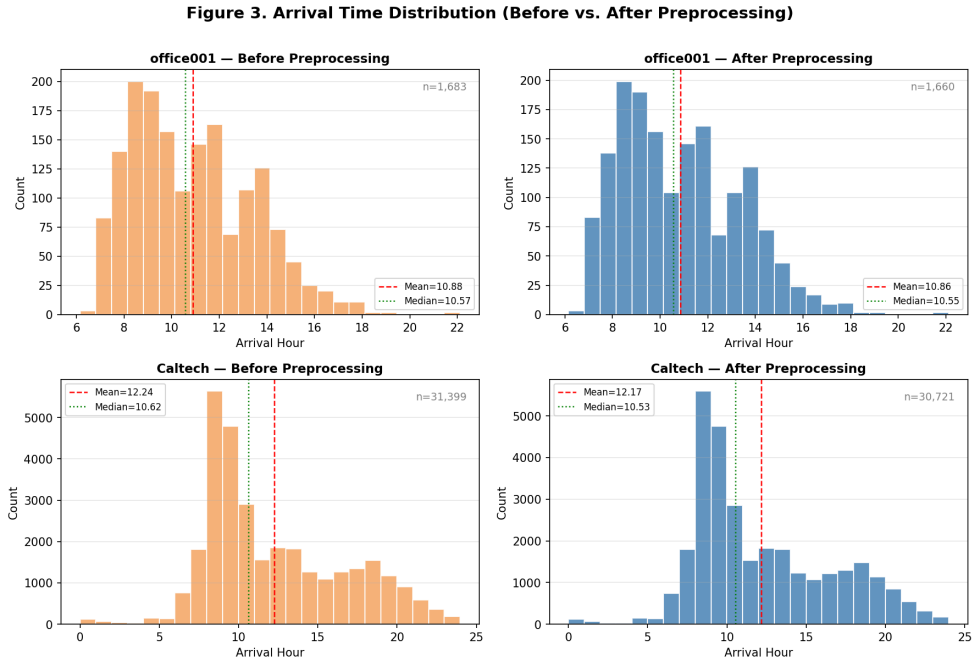

Figure 3.3: Arrival Time Distribution before and after preprocessing for office001 and Caltech. Both sites show charging activity concentrated during daytime hours, consistent with workplace charging patterns. Similarly, the overall distribution shape is preserved after preprocessing for both sites.

3.3 Instance Generation

Each scheduling instance corresponds to a single day of charging sessions. For a given target date, all sessions with `connection_time` falling on that date are extracted and assigned to the MIP parameters $(a_i, d_i, e_i, \bar{r}_i)$. Sessions in which the maximum possible energy $\bar{r}_i \cdot (d_i - a_i + 1) \cdot \Delta t < e_i$ are removed as physically infeasible.

Two instance scales are used in the experiments:

- **Small-scale (office001)**: 1–14 vehicles per day (mean: 3.1), used for model validation and small-scale analysis.
- **Large-scale (Caltech)**: 1–105 vehicles per day (mean: 28.4), used for full-scale experiments and scalability analysis.

3.4 Evaluation Metrics

The following metrics are used to compare the four methods (MIP, UC, EDF, RR):

Table 3.4: Evaluation Metrics

Metric	Symbol	Unit	Description
Peak power	Z^*	kW	Maximum total power across all time slots
Peak reduction rate	ρ	%	$(Z_{baseline} - Z^*)/Z_{baseline} \times 100$
Energy fulfillment rate	η	%	Fraction of vehicles fully charged
Computation time	T_{solve}	s	Wall-clock solver time

4 Results and Analysis

4.1 Small-Scale Validation (office001)

4.1.1 MIP Optimal Solution

The MIP model is first validated on the office001 dataset, which contains 1,600 sessions that span 543 scheduling days with an average of 3.1 vehicles per day (maximum: 14 vehicles). This small-scale setting serves as a sanity check to verify that the model produces feasible and optimal solutions before scaling to the larger Caltech dataset.

Table 4.1 summarizes the MIP solution statistics for 543 days.

Table 4.1: MIP Solution Summary - office001 (543 days)

Metric	Mean	Std	Min	Max
Peak Power Z^* (kW)	5.45	2.95	0.40	19.08
Energy Fulfillment Rate (%)	100.0	0.0	100.0	100.0
Computation Time (s)	0.003	0.001	0.001	0.005

The MIP model achieves 100% energy fulfillment across all 543 instances, confirming that the formulation is feasible and correctly enforces the charging completion constraint (C1). The computation time remains below 5 milliseconds for all instances, reflecting the small problem size (maximum 14 vehicles $\times$ 96 slots). Figure 4.1 illustrates the relationship between vehicle count and computation time.

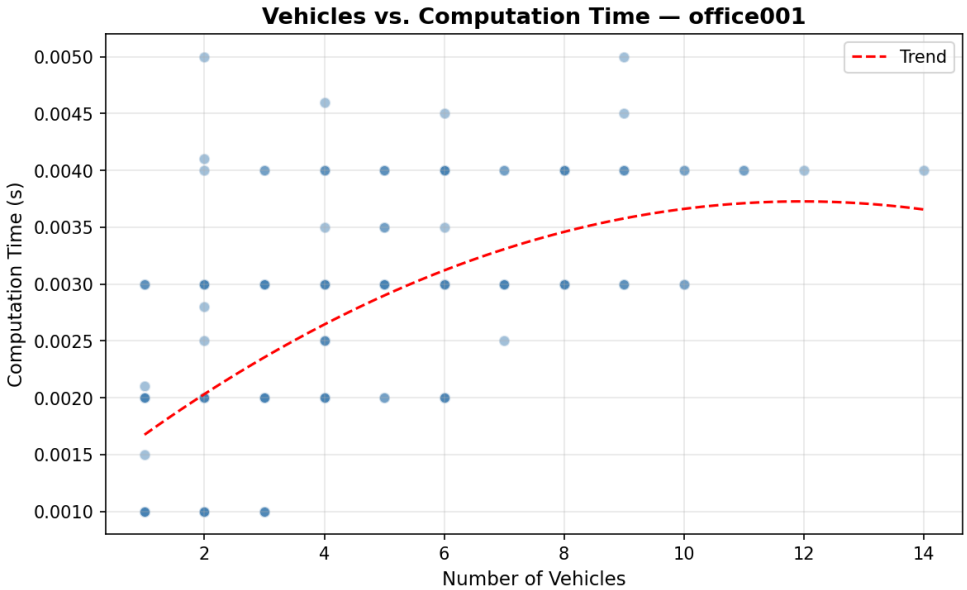

Figure 4.1: Vehicles vs. computation time for office001. Computation time remains below 5 ms across all instance sizes, confirming computational tractability at this scale.

4.1.2 Comparison with Baselines

Table 4.2 compares the peak power statistics of all four methods throughout the entire 543-day period.

Table 4.2: Peak Power Comparison — office001 (543 days)

Method	Mean (kW)	Std (kW)	Min (kW)	Max (kW)	Fulfillment	Reduction vs. UC
MIP	5.45	2.95	0.40	19.08	100%	56.5%
UC	13.07	7.30	3.04	43.20	100%	—
EDF	13.07	7.30	3.04	43.20	100%	0.0%
RR	13.07	7.30	3.04	43.20	100%	0.0%

The MIP model achieves a mean peak reduction of **56.5%** relative to uncontrolled charging. In particular, UC, EDF, and RR produce identical results on all days. This is attributable to the characteristics of the office001 site: with an average of only 3.1 vehicles per day and a transformer capacity of 50 kW, the concurrent charging demand rarely approaches the capacity limit. As a result, all three heuristics operate effectively without binding constraints, charging each vehicle at its maximum rate upon connection.

Figure 4.2 presents the daily peak power trends for all four methods throughout the period.

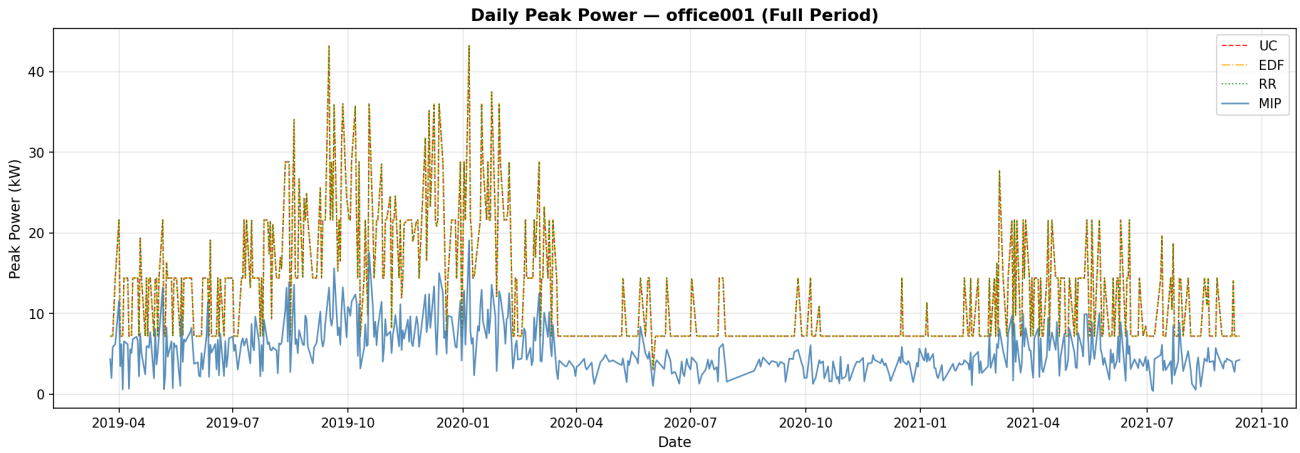

Figure 4.2: Daily peak power over the full office001 period (2019-03-25 to 2021-09-13). UC, EDF, and RR overlap entirely throughout the period. A notable decline in peak power is observed after April 2020, likely attributable to reduced vehicle occupancy during the COVID-19 pandemic.

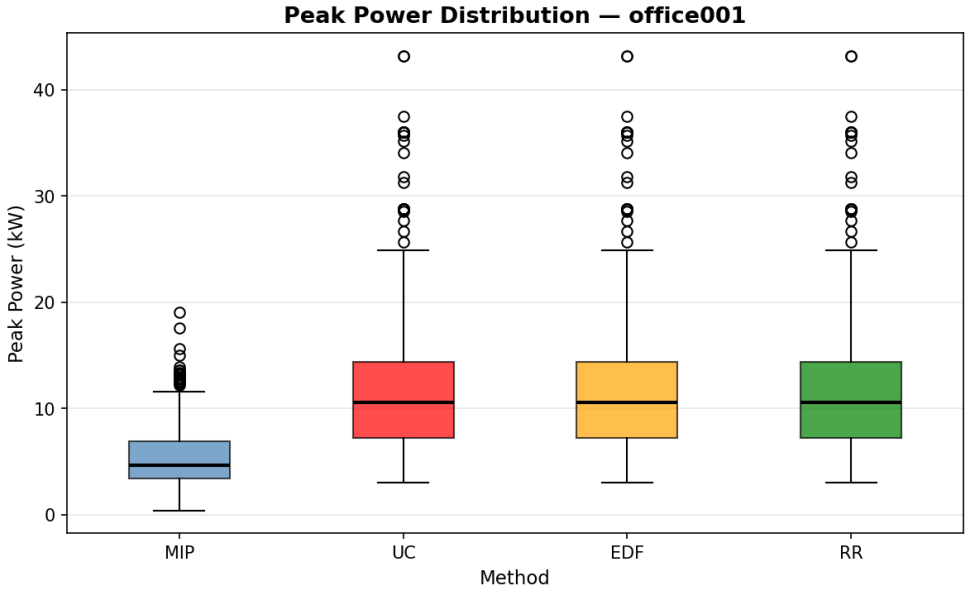

Figure 4.3: Peak power distribution across all methods — office001. The MIP box is substantially lower and narrower than the heuristic methods, indicating both lower and more stable peak power.

Figure 4.4 shows the distribution of the daily peak reduction rates achieved by the MIP model relative to UC.

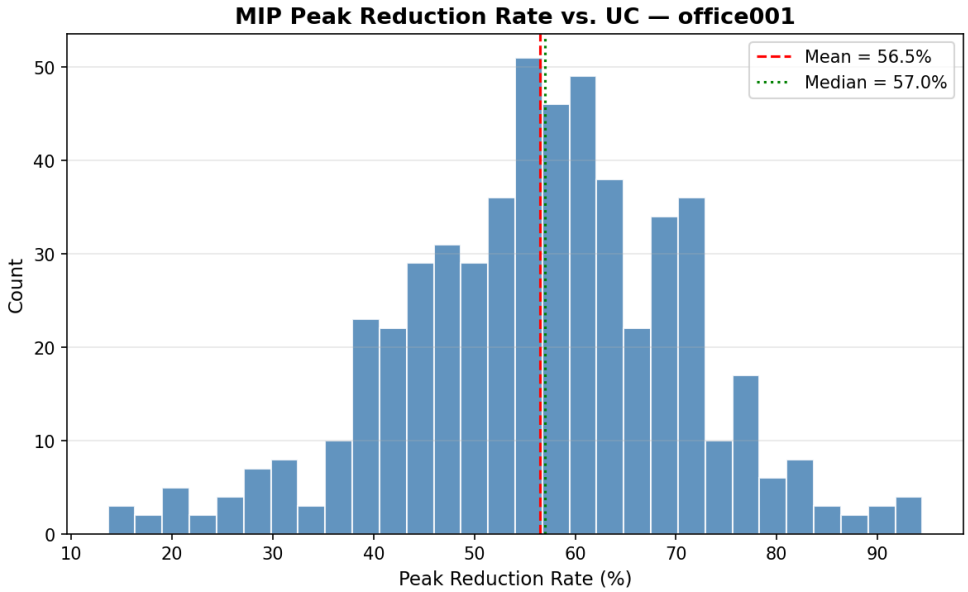

Figure 4.4: Distribution of MIP peak reduction rate vs. UC — office001. The reduction rate is concentrated between 40% and 80%, with a mean of 56.5% and median of 57.0%, indicating consistent performance across varying instance sizes.

4.2 Large-Scale Experiment (Caltech)

4.2.1 MIP Optimal Solution

The MIP model is applied to the full Caltech dataset, comprising 30,721 sessions on 1,081 scheduling days with an average of 28.4 vehicles per day (maximum: 105 vehicles). This large-scale experiment evaluates both solution quality and computational scalability under realistic charging conditions.

Table 4.3 summarizes the MIP solution statistics for 1,081 days.

Table 4.3: MIP Solution Summary — Caltech (1,081 days)

Metric	Mean	Std	Min	Max
Peak Power Z^* (kW)	16.80	12.71	0.18	56.59
Energy Fulfillment Rate (%)	100.0	0.0	100.0	100.0
Computation Time (s)	0.009	0.018	0.001	0.322

Figure 4.5 illustrates the relationship between vehicle count and MIP computation time across all instances.

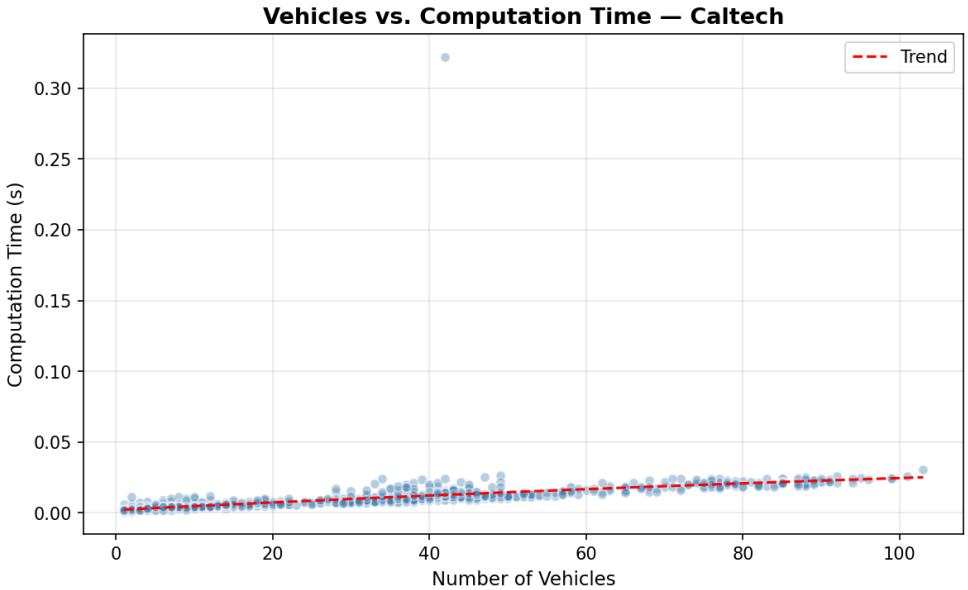

Figure 4.5: Vehicles vs. computation time for Caltech. Computation time increases gradually with vehicle count but remains below 0.05 seconds for 99% of instances, confirming that the MIP formulation is computationally tractable even at large scale.

4.2.2 Comparison with Baselines

Table 4.4 compares the peak power statistics of all four methods.

Table 4.4: Peak Power Comparison — Caltech (1,081 days)

Method	Mean (kW)	Std (kW)	Min (kW)	Max (kW)	Fulfillment	Reduction vs. UC
MIP	16.80	12.71	0.18	56.59	100%	75.9%
UC	74.57	60.33	3.00	327.52	100%	—
EDF	69.33	49.21	3.00	150.00	100%	2.4%
RR	65.10	43.62	3.00	150.00	100%	5.4%

The MIP model achieves a mean peak reduction of **75.9%** compared to uncontrolled charging, with a median of 77.4%, demonstrating consistent and reliable performance throughout all 1,081 days. In contrast, EDF and RR achieve only 2.4% and 5.4% mean reduction respectively, with median reductions of 0%. This indicates that rule-based heuristics provide negligible benefit over uncontrolled charging at this scale, where the large number of simultaneously connected vehicles creates complex scheduling interactions that simple priority rules cannot effectively resolve.

Furthermore, the MIP model exhibits the lowest standard deviation (12.71 kW) among all methods, indicating not only a lower but also a more stable peak power under varying daily conditions. The max-

imum peak of UC reaches 327.52 kW-exceeding the transformer capacity of 150 kW-while EDF and RR are bounded at 150 kW due to the transformer capacity constraint applied in their implementation.

Figure 4.6 presents the daily peak power trends over the entire period.

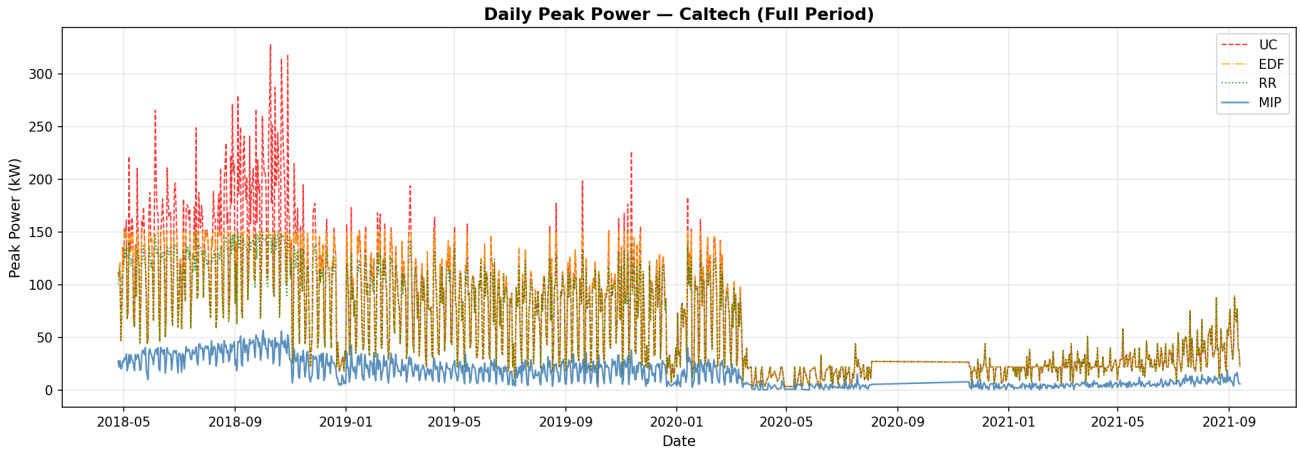

Figure 4.6: Daily peak power over the full Caltech period (2018-04-25 to 2021-09-13). UC reaches up to 325 kW during peak periods, while MIP consistently maintains low peak power. A sharp decline in all methods is observed after April 2020, consistent with reduced campus occupancy during the COVID-19 pandemic.

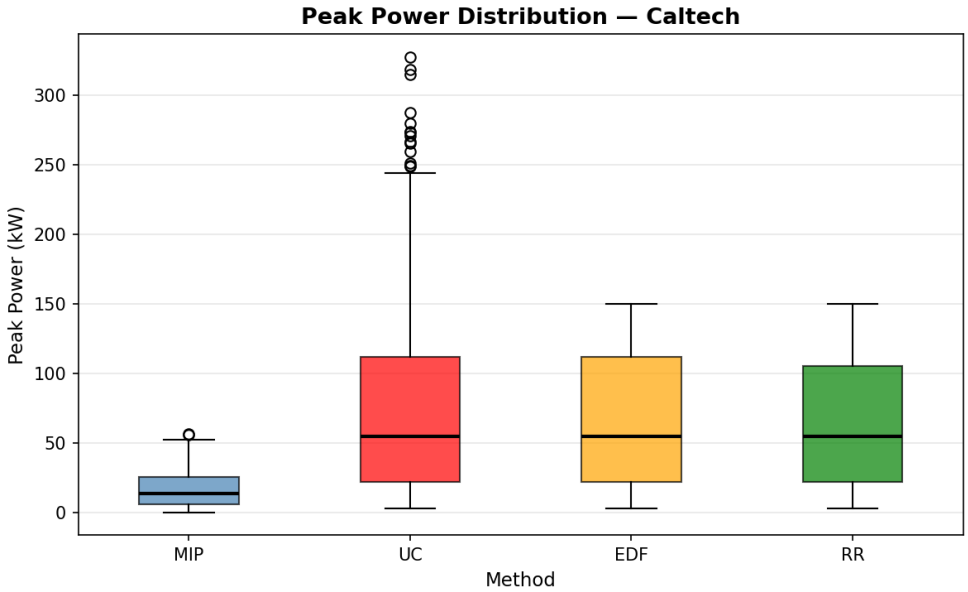

Figure 4.7: Peak power distribution across all methods — Caltech. The MIP inter-quartile range is substantially narrower than the heuristic methods, indicating not only lower but also more stable peak power.

Figure 4.8 shows the distribution of peak reduction rates for all three methods relative to UC.

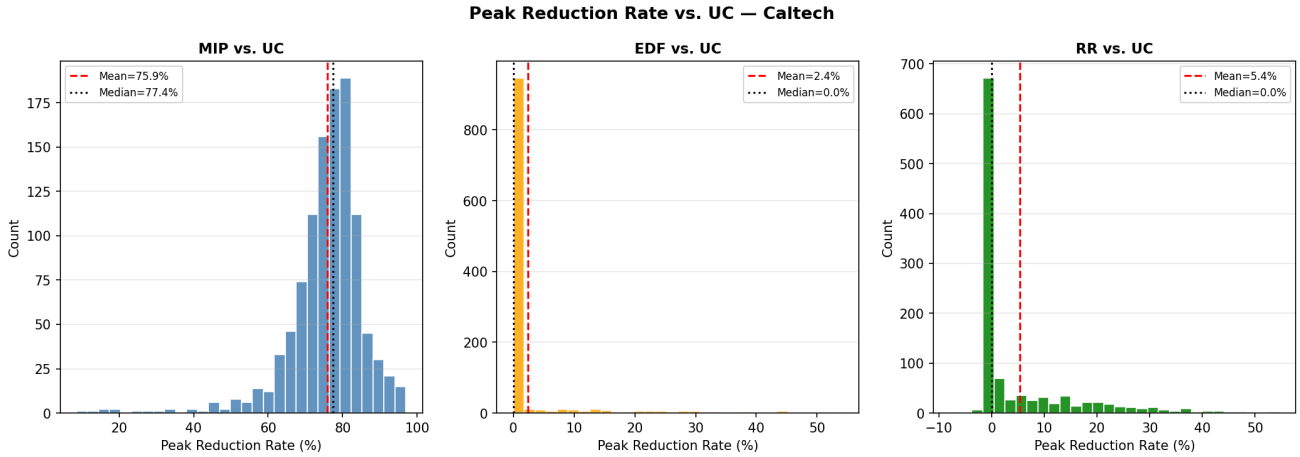

Figure 4.8: Peak reduction rate distributions vs. UC — Caltech. MIP achieves consistent reductions of 60–90% across most days. EDF and RR show near-zero median reductions, with RR occasionally producing negative reduction rates (i.e., higher peak than UC) on days where uniform load distribution is suboptimal.

The EDF vs. RR scatter plot (Figure 4.9) reveals that RR generally achieves a lower peak power than EDF under high-load conditions, while the two methods perform equivalently under low-load conditions.

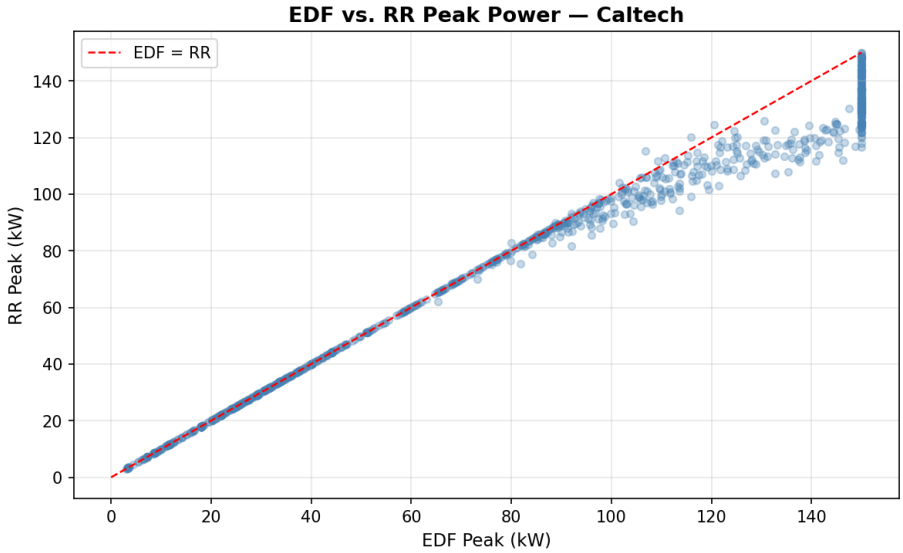

Figure 4.9: EDF vs. RR peak power scatter plot — Caltech. Points below the diagonal (EDF = RR) indicate days where RR achieves lower peak than EDF. This pattern becomes more pronounced under high-load conditions where transformer capacity is a binding constraint.

4.2.3 Sensitivity Analysis

To assess the robustness of the MIP model with respect to the transformer capacity constraint, a sensitivity analysis is conducted by varying the transformer capacity C over a range of values for three representative instances of different scales: a small instance (10 vehicles, 2019-07-21), a medium instance (40 vehicles, 2019-08-29), and a large instance (80 vehicles, 2018-10-25).

Figure 4.10 presents the MIP peak power and energy fulfillment rate as a function of C for each instance.

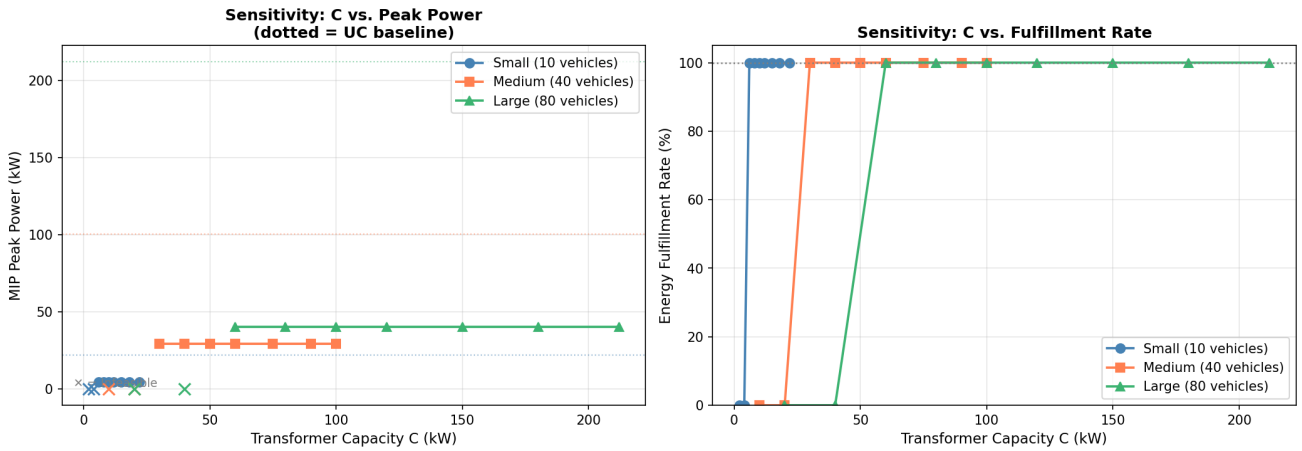

Figure 4.10: Sensitivity analysis of transformer capacity C on MIP peak power (left) and energy fulfillment rate (right) for three representative instances. Shaded regions indicate infeasible settings; dotted horizontal lines indicate the UC baseline peak.

Two key observations emerge from the analysis.

Feasibility threshold. For each instance, a minimum transformer capacity threshold exists below which the scheduling problem becomes infeasible — that is, no schedule can deliver the required energy to all vehicles within their stay windows under the given capacity constraint. This threshold increases with the number of vehicles: approximately 6 kW for the small instance, 25 kW for the medium instance, and 60 kW for the large instance. Below these thresholds, the MIP solver returns an infeasible status, confirming that a minimum power budget is necessary to guarantee full energy delivery.

Non-binding capacity above the threshold. Once C exceeds the feasibility threshold, the optimal peak power of the MIP converges immediately and remains stable regardless of further capacity increases. This indicates that the transformer capacity constraint (C5) becomes non-binding above the threshold, and the achieved peak is determined solely by the energy requirements and stay durations of the vehicles — not by the imposed capacity limit. In all feasible cases, the MIP peak remains well below the UC baseline (dotted lines), confirming that coordinated scheduling achieves significant peak reduction independently of the transformer capacity setting.

These findings suggest that the MIP-based scheduler is robust to variations in transformer capacity above the feasibility threshold and that the actual transformer capacity of the Caltech site (150 kW) provides sufficient headroom for optimal scheduling across all observed instance sizes.

4.3 Extension Experiments (Optional)

The following extension experiments are conducted exclusively on the Caltech dataset, as the office001 site operates at a scale where transformer capacity is rarely a binding constraint, making the effect of optional constraints negligible. Three representative instances (Small: 10 vehicles, Medium: 40 vehicles, Large: 80 vehicles) are used consistently across all extension experiments.

4.3.1 Effect of C7

The minimum charging rate constraint (C7) is evaluated by varying $\underline{r}_i$ from 0 (no constraint) to 4.0 kW in three representative instances. Figure 4.11 presents the effect on peak power, energy fulfillment rate, and average charging interruptions per vehicle.

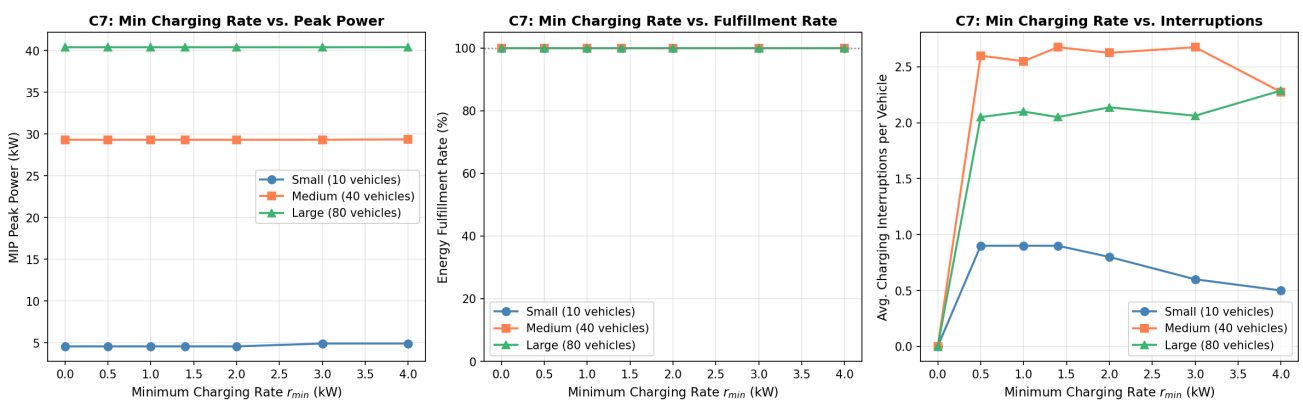

Figure 4.11: Effect of minimum charging rate constraint (C7) on peak power (left), energy fulfillment rate (center), and average charging interruptions per vehicle (right) for three representative Caltech instances.

Two observations emerge from the results. First, the maximum MIP power and the energy fulfillment rate remain unchanged in all values of $\underline{r}_i$, indicating that C7 does not affect the quality of the solution within the tested range. This is because the optimal peak is determined by the total energy demand and the duration of stay of the vehicle, which are independent of the minimum charging rate threshold.

Second, the average number of charging interruptions per vehicle increases sharply when $\underline{r}_i > 0$, and remains elevated across all values tested. This is a direct consequence of C7: when a charger is active, it must deliver at least $\underline{r}_i$ kW, which reduces the solver's flexibility to assign small fractional rates.

As a result, the solver more frequently switches the binary variable y_{it} on and off, leading to more interruptions.

Three findings suggest that while C7 does not compromise peak minimization or charging completion, it introduces a practical trade-off: enforcing a minimum charging rate increases charging interruption, which may negatively affect battery health and user experience. Based on these results, C7 is excluded from the main experimental configuration.

4.3.2 Effect of C8

The limitation of the charging interruption limit (C8) is evaluated by varying the maximum number of interruptions allowed K_i from 0 (no interruptions allowed) to unlimited ($K_i = \infty$, equivalent to the base MIP without C8). Figure 4.12 presents the results.

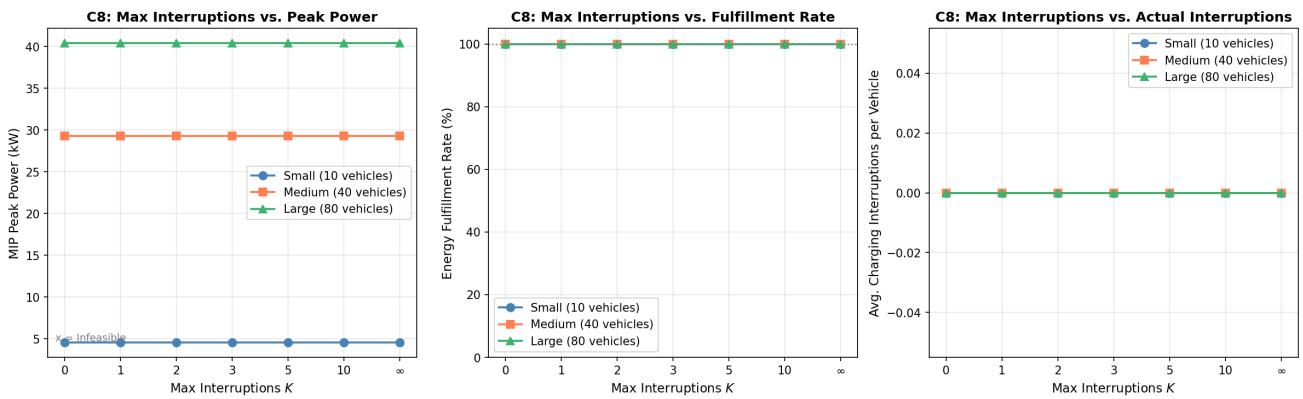

Figure 4.12: Effect of charging interruption limit (C8) on peak power (left), energy fulfillment rate (center), and actual average interruptions per vehicle (right).

Three observations emerge from the results. First, when $K_i = 0$ (no interruptions allowed), the small instance (10 vehicles) becomes infeasible. This indicates that for certain vehicles with short stay durations, a single continuous charging session is insufficient to deliver the required energy-at least one interruption is necessary to accommodate the peak minimization objective.

Second, for $K_i \geq 1$, the maximum power and the energy fulfillment rate remain unchanged across all values of K_i, indicating that C8 is non-binding above $K_i = 1$.

Third, the actual average interruption count per vehicle is effectively zero in all instances and in all values of K_i. This is explained by the structure of the base MIP: since the objective minimizes peak power, the solver naturally distributes charging load evenly across time slots, which inherently favors continuous charging sessions with minimal on/off toggling. As a result, the base MIP already produces schedules with near-zero interruptions, rendering C8 non-binding in practice.

These findings suggest that C8 provides no meaningful benefit in this setting and is therefore excluded from the main experimental configuration. Furthermore, the near-zero interruption count of the base

MIP solution indicates that the resulting schedules are practically favorable from a battery health and user experience perspective, even without explicitly enforcing C8.

4.3.3 TOU Cost Integration

To explore the trade-off between peak power minimization and TOU energy cost, the objective function is extended to incorporate both components:

$$\min \sum_{t \in T} \left(\lambda_t \cdot \sum_{i \in I} r_{it} \cdot \Delta t \right) + \mu \cdot Z \tag{4.1}$$

where λ_t denotes the price of TOU electricity that varies over time (SCE TOU-EV-8 tariff) and μ is the demand charge coefficient (\$/kW). The demand charge coefficient μ ranges from 0 (TOU energy cost only) to 20 \$/kW in three representative instances. Figure 4.13 presents the results.

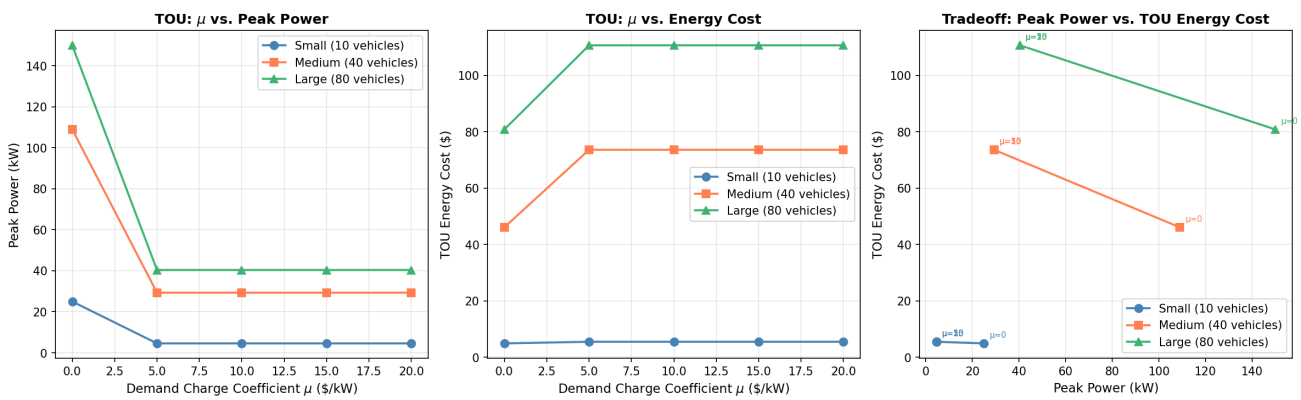

Figure 4.13: Effect of demand charge coefficient μ on peak power (left), TOU energy cost (center), and the resulting peak-cost tradeoff curve (right) for three representative Caltech instances.

Three key observations emerge from the analysis.

The reduction in peak with increasing μ. When $\mu = 0$, the objective reduces to minimization of the total energy cost of the TOU, which concentrates charging during off-peak periods regardless of concurrent demand. This results in high peak power: 25 kW for the small instance, 109 kW for the medium instance, and 150 kW for the large instance. As μ increases to 5 \$/kW, peak power drops sharply to levels comparable to the base MIP (min Z) solution — 4.6 kW, 29 kW, and 40 kW respectively. Beyond $\mu = 5$ \$/kW, peak power converges and no further reduction is observed.

The energy cost of TOU increases with μ. As the demand charge coefficient increases, the solver shifts the load away from high-demand periods to reduce the peak power, which inevitably requires utilizing higher-priced time slots. Consequently, the TOU energy cost rises sharply between $\mu = 0$ and $\mu = 5$, then stabilizes as the schedule converges to the peak-minimizing solution.

Clear peak-cost tradeoff. The rightmost panel illustrates the tradeoff between peak power and TOU energy cost. Moving from $\mu = 0$ to $\mu = 20$, the peak power decreases while the energy cost increases, forming a clear Pareto-like frontier. This tradeoff is most pronounced for the large instance (80 vehicles), where the gap between pure cost minimization ($\mu = 0$) and pure peak minimization ($\mu = \infty$) is largest.

These findings confirm that a small demand charge coefficient ($\mu \geq 5$ \$/kW) is sufficient to achieve near-optimal peak reduction, while $\mu = 0$ results in unacceptably high peak demand despite minimizing energy cost. The choice of μ therefore represents a practical tuning parameter for operators seeking to balance electricity cost and grid stability.

5 Conclusion

5.1 Key Findings

This study formulates the EV charging scheduling problem as a Mixed-Integer Programming (MIP) model and evaluates its performance on real-world data from two Caltech ACN sites over multi-year periods. The following key findings are reported.

F1. Consistent and substantial peak reduction. The MIP-based scheduler achieves mean peak reductions of 56.5% on office001 and 75.9% on Caltech relative to uncontrolled charging, with median reductions of 57.0% and 77.4% respectively. These results are consistent across the full dataset periods (543 days for office001, 1,081 days for Caltech), demonstrating reliable performance under varying daily conditions.

F2. Heuristic methods provide negligible benefit at scale. On office001, UC, EDF, and RR produce identical results due to the small number of vehicles and large transformer headroom. On Caltech, EDF and RR achieve only 2.4% and 5.4% mean peak reduction respectively, with median reductions of 0%. This confirms that rule-based heuristics are insufficient for large-scale coordinated charging.

F3. Computational tractability. The MIP model solves to optimality within 0.005 seconds on office001 and within 0.35 seconds on Caltech (up to 105 vehicles), confirming its suitability for real-time or day-ahead scheduling deployment.

F4. Transformer capacity threshold. The sensitivity analysis reveals a minimum feasibility threshold for transformer capacity below which no feasible schedule exists. Above this threshold, the MIP peak converges immediately and remains stable, indicating that the Caltech transformer capacity (150 kW) provides sufficient headroom for optimal scheduling across all observed instance sizes.

F5. Optional constraints are non-binding. C7 (minimum charging rate) does not affect peak or fulfillment but increases charging interruptions, while C8 (interruption limit) is already satisfied by the base MIP without explicit enforcement. Both constraints are therefore excluded from the main configuration.

F6. TOU-peak tradeoff. Incorporating a demand charge coefficient $\mu \geq 5$ \$/kW into the objective function is sufficient to achieve near-optimal peak reduction, while $\mu = 0$ results in high peak demand despite minimizing TOU energy cost. The value of μ serves as a practical tuning parameter for operators balancing electricity cost and grid stability.

5.2 Insights

There are four Insights in this project.

I1. Scale-dependent effectiveness. The benefit of MIP-based optimization grows with instance scale. On office001, where the transformer capacity is rarely a binding constraint, even simple heuristics perform adequately. On Caltech, where simultaneous charging demand regularly approaches transformer limits, only the MIP model achieves meaningful peak reduction. This suggests that coordinated scheduling is most valuable in high-density charging environments.

I2. MIP naturally produces interruption-free schedules. The near-zero charging interruption count observed in the base MIP solution — without any explicit C8 constraint — indicates that peak minimization inherently encourages continuous charging. This is a practically favorable property, as uninterrupted charging is generally preferred for battery health and user experience.

I3. Real-world data captures operational variability. The use of real ACN session data reveals patterns that synthetic benchmarks cannot capture, including the sharp decline in charging activity during the COVID-19 pandemic (post April 2020) and day-to-day variability in vehicle counts and energy demands. These patterns directly affect scheduling difficulty and solution quality, underscoring the value of data-driven evaluation.

I4. Practical deployment readiness. The MIP model's sub-second computation time, 100% energy fulfillment rate, and consistent peak reduction performance across over 1,600 scheduling instances suggest that it is ready for practical deployment as a day-ahead or real-time EV charging scheduler.

5.3 Limitations and Future Work

The current study operates under a deterministic offline setting, assuming full knowledge of vehicle arrival times, departure times, and energy requirements. In practice, these parameters are uncertain and must be estimated or predicted. Extending the formulation to handle stochastic or online settings — where scheduling decisions are made as vehicles arrive — represents a natural direction for future work.

Additionally, the objective is currently limited to peak power minimization. Jointly optimizing TOU energy cost and peak demand under realistic SCE commercial tariffs (e.g., TOU-GS-2 with non-zero demand charges) would provide a more complete cost model. The sensitivity analysis in Section 4.3.3 provides a preliminary foundation for this extension.

Finally, the model assumes a single transformer serving all chargers. Incorporating hierarchical network constraints — as reflected in the ACN's multi-level constraint structure (CC Pod, AV Pod, Secondary, Primary) — would more accurately represent the physical infrastructure and may reveal additional scheduling opportunities.

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